

Speech Recognition using Mel-Frequency Cepstral Coefficients (MFCCs) *Part B*

Lab 13

Department of Artificial Intelligence
NUML — Islamabad

Submitted by: **Junaid Asif**
Roll No: **BSAI-144**
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Instructor: Dr. Sajjad Ghauri

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1 Introduction

In previous labs, speech recognition was performed using aggregated MFCC features and classical classifiers such as KNN and SVM. While effective for isolated words, aggregation removes temporal information from speech signals.

This lab introduces **Dynamic Time Warping (DTW)**, a sequence-based classification technique that compares entire MFCC sequences directly, preserving temporal dynamics. Additionally, conceptual extensions toward Hidden Markov Models (HMMs) and deep learning-based speech recognition are discussed.

2 Objective

- To understand Dynamic Time Warping for sequence alignment.
- To implement DTW-based comparison using MFCC sequences.
- To conceptualize DTW-based classification.
- To introduce advanced speech recognition techniques such as HMMs and deep learning.

3 Lab Task 1: Sequence-Based Classification using DTW

3.1 Understanding Dynamic Time Warping (DTW)

Dynamic Time Warping is an algorithm used to align two sequences of varying lengths by minimizing the cumulative distance between them. Unlike simple distance measures, DTW allows non-linear alignment in time, making it highly suitable for speech signals that vary in speaking rate and duration.

DTW is particularly useful in speech recognition because:

- Speakers pronounce the same word at different speeds.
- Utterances vary in length.
- Temporal variations should not penalize similarity excessively.

3.2 DTW-Based Word Comparison

Two recordings of the same word typically yield a smaller DTW distance compared to recordings of different words, indicating higher similarity.

3.3 MATLAB Code: DTW Implementation

```
1 % lab_task_3.m
2 clear; clc; close all;
3
4 audioFolderPath = 'audio/';
```

```

5 n_mfcc_coeffs = 13;
6
7 file1 = 'yes_1.wav';
8 file2 = 'yes_2.wav';
9 file3 = 'no_1.wav';
10
11 [y1, fs] = audioread(fullfile(audioFolderPath, file1));
12 [y2, ~] = audioread(fullfile(audioFolderPath, file2));
13 [y3, ~] = audioread(fullfile(audioFolderPath, file3));
14
15 y1 = mean(y1,2); y2 = mean(y2,2); y3 = mean(y3,2);
16 y1 = y1/max(abs(y1)); y2 = y2/max(abs(y2)); y3 = y3/max(abs(y3));
17
18 afe = audioFeatureExtractor( ...
19     'SampleRate', fs, ...
20     'Window', hann(round(0.025*fs),'periodic'), ...
21     'OverlapLength', round(0.010*fs), ...
22     'mfcc', true, ...
23     'NumCoefficients', n_mfcc_coeffs);
24
25 mfccs1 = extract(afe, y1);
26 mfccs2 = extract(afe, y2);
27 mfccs3 = extract(afe, y3);
28
29 mfccs1 = (mfccs1-mean(mfccs1,1))./(std(mfccs1,0,1)+eps);
30 mfccs2 = (mfccs2-mean(mfccs2,1))./(std(mfccs2,0,1)+eps);
31 mfccs3 = (mfccs3-mean(mfccs3,1))./(std(mfccs3,0,1)+eps);
32
33 dtw_same = dtw_distance(mfccs1, mfccs2);
34 dtw_diff = dtw_distance(mfccs1, mfccs3);
35
36 fprintf('DTW distance (same word): %.4f\n', dtw_same);
37 fprintf('DTW distance (different word): %.4f\n', dtw_diff);

```

3.4 DTW Distance Function

```

1 function dist = dtw_distance(s, t)
2 n = size(s,1);
3 m = size(t,1);
4
5 D = inf(n+1, m+1);
6 D(1,1) = 0;
7
8 for i = 1:n
9     for j = 1:m
10         cost = norm(s(i,:) - t(j,:));
11         D(i+1,j+1) = cost + min([D(i,j+1), D(i+1,j), D(i,j)]);
12     end
13 end
14
15 dist = D(n+1,m+1);
16 end

```

3.5 DTW-Based Classifier (Conceptual)

A DTW-based classifier compares a test utterance against all training utterances and assigns the label corresponding to the minimum DTW distance. Although accurate for small datasets, this approach becomes computationally expensive for large vocabularies.

4 Questions for Lab Task 1

How does DTW distance reflect similarity? Lower DTW distance indicates higher similarity between two speech utterances, as less warping is required to align them.

Computational challenges of DTW: DTW has high computational cost $O(NM)$, making it impractical for large datasets without optimization.

When is DTW preferred over aggregation? DTW is preferred when temporal structure is important and speech duration varies significantly.

5 Lab Task 2: Enhancing the System (Conceptual)

5.1 Hidden Markov Models (HMMs)

Simple classifiers and DTW struggle with continuous speech and large vocabularies. Hidden Markov Models address this by modeling speech as a sequence of hidden states with probabilistic transitions.

Key components of HMMs:

- States
- Observations
- Transition probabilities
- Emission probabilities

Problems solved by HMMs:

- Evaluation (Forward Algorithm)
- Decoding (Viterbi Algorithm)
- Learning (Baum-Welch Algorithm)

5.2 HMM-Based ASR Pipeline

In ASR systems, HMMs model phonemes or words, while language models help predict word sequences.

5.3 Deep Learning for Speech Recognition

Modern ASR systems use deep learning models such as RNNs, LSTMs, and Transformers. These models learn temporal patterns directly from MFCC sequences and outperform traditional methods.

6 Questions for Lab Task 2

How do HMMs handle variability better than DTW? HMMs model duration and transitions probabilistically, making them robust to variations in speech.

Challenges in real ASR systems: Large datasets, noise robustness, computational cost, and real-time constraints.

Why deep learning dominates modern ASR? Deep models automatically learn complex temporal and spectral patterns, achieving higher accuracy.

Future extensions: Incorporating delta and delta-delta features, larger datasets, deep neural networks, and end-to-end ASR systems.

7 Conclusion

This lab introduced sequence-based speech recognition using DTW and explored advanced techniques such as HMMs and deep learning. DTW preserves temporal information but suffers from scalability issues, motivating the use of probabilistic and deep learning-based approaches in modern speech recognition systems.